



Lecture 11

Module 9: Cloud Architecture



Module overview

Topics

- AWS Well-Architected Framework
- Reliability and high availability
- AWS Trusted Advisor



Knowledge check



Module objectives

After completing this module, you should be able to:

- Describe the AWS Well-Architected Framework, including the six pillars
- Identify the design principles of the AWS Well-Architected Framework
- Explain the importance of reliability and high availability
- Identify how AWS Trusted Advisor helps customers
- Interpret AWS Trusted Advisor recommendations

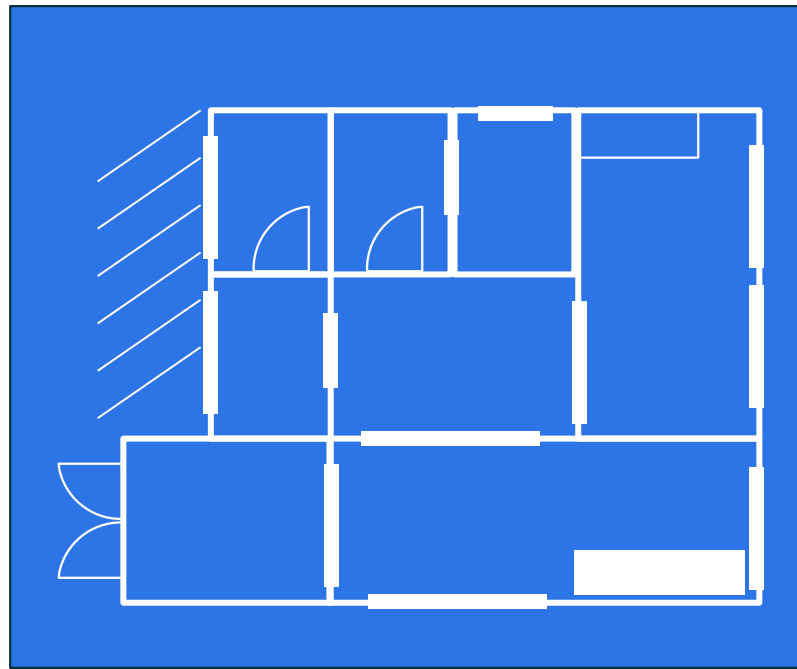


Section 1: AWS Well-Architected Framework

Module 9: Cloud Architecture



Architecture: designing and building



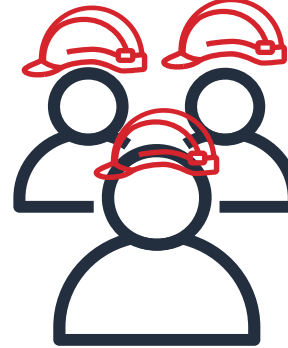
Structure design



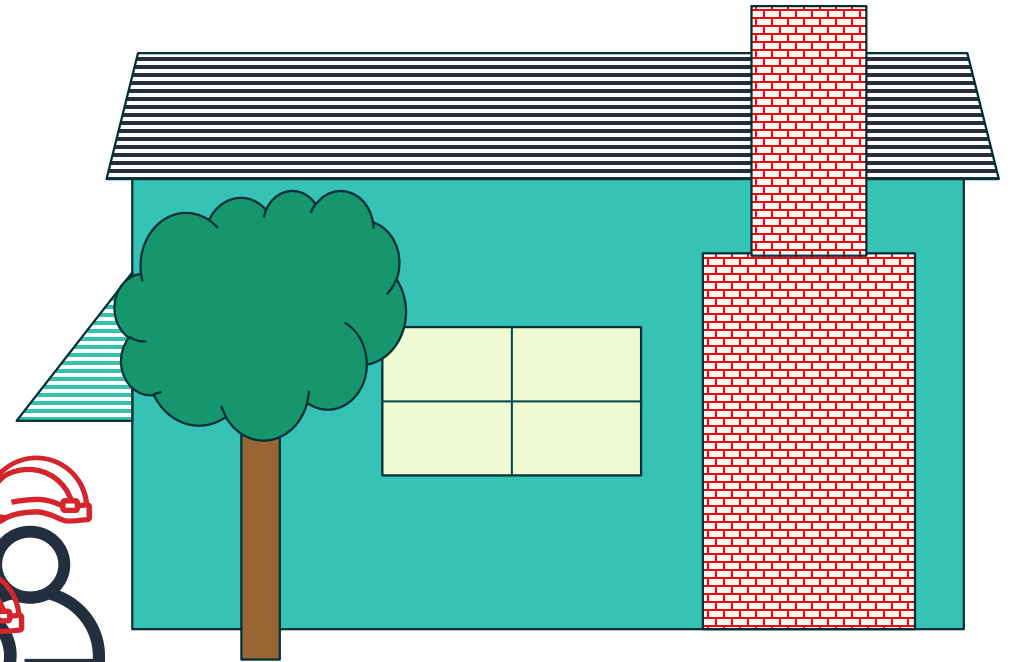
Customer
(Decision maker)



Architect



Building crew
(Delivery team)



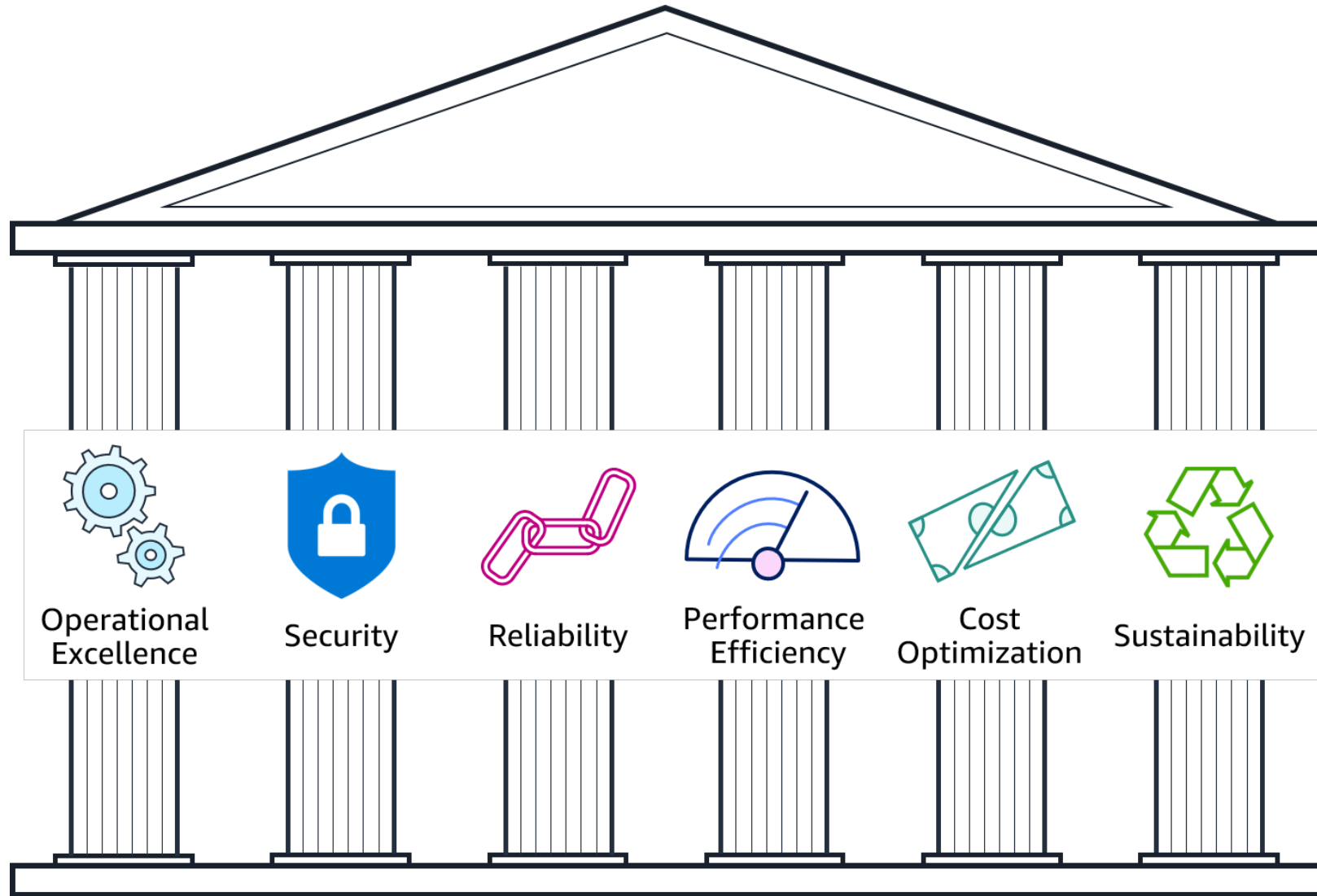
Completed structure



What is the AWS Well-Architected Framework?

- A guide for designing infrastructures that are:
 - ✓ Secure
 - ✓ High-performing
 - ✓ Resilient
 - ✓ Efficient
- A consistent approach to evaluating and implementing cloud architectures
- A way to provide best practices that were developed through lessons learned by reviewing customer architectures

Pillars of the AWS Well-Architected Framework





Pillar organization

Best practice area

Question text

Question context

Best practices

Identity and Access Management

SEC 1: How do you manage credentials and authentication?

Credential and authentication mechanisms include passwords, tokens, and keys that grant access directly or indirectly in your workload. Protect credentials with appropriate mechanisms to help reduce the risk of accidental or malicious use.

Best practices:

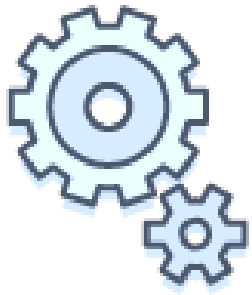
- Define requirements for identity and access management
- Secure AWS account root user
- Enforce use of multi-factor authentication
- Automate enforcement of access controls
- Integrate with centralized federation provider
- Enforce password requirements
- Rotate credentials regularly
- Audit credentials periodically



Operational Excellence pillar

Operational Excellence pillar – deliver business value

Operational Excellence pillar

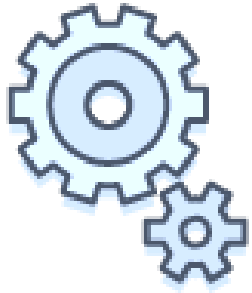


Deliver
business
value

- **Focus**
 - Run and monitor systems to deliver business value, and to continually improve supporting processes and procedures.
- **Key topics**
 - Automating changes
 - Responding to events
 - Defining standards to manage daily operations

Operational excellence design principles

Operational Excellence pillar



Deliver
business
value

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operational events and failures



Operational excellence questions

Organization

- How do you determine what your priorities are?
- How do you structure your organization to support your business outcomes?
- How does your organizational culture support your business outcomes?

Prepare

- How do you design your workload so that you can understand its state?
- How do you reduce defects, ease remediation, and improve flow into production?
- How do you mitigate deployment risks?
- How do you know that you are ready to support a workload?

Operate

- How do you understand the health of your workload?
- How do you understand the health of your operations?
- How do you manage workload and operations events?

Evolve

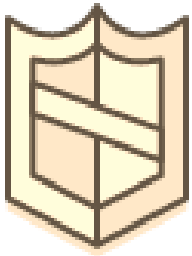
- How do you evolve operations?



Security pillar

Security pillar – protect and monitor systems

Security pillar



Protect and
monitor
systems

- **Focus**

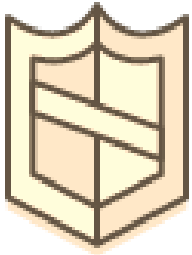
- Protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies.

- **Key topics**

- Protecting confidentiality and integrity of data
- Identifying and managing who can do what
- Protecting systems
- Establishing controls to detect security events

Security design principles

Security pillar



Protect and
monitor
systems

- Implement a strong identity foundation
- Enable traceability
- Apply security at all layers
- Automate security best practices
- Protect data in transit and at rest
- Keep people away from data
- Prepare for security events



Security questions

Security

- How do you securely operate your workload?

Identity and access management

- How do you manage identities for people and machines?
- How do you manage permissions for people and machines?

Detection

- How do you detect and investigate security events?

Infrastructure protection

- How do you protect your network resources?
- How do you protect your compute resources?

Data protection

- How do you classify your data?
- How do you protect your data at rest?
- How do you protect your data in transit?

Incident response

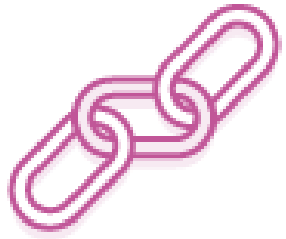
- How do you anticipate, respond to, and recover from incidents?



Reliability pillar

Reliability pillar – recover from failure and mitigate disruption

Reliability pillar

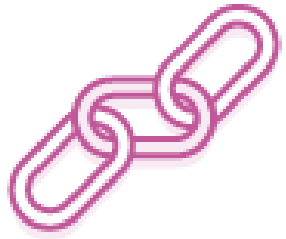


Recover from
failure and
mitigate
disruption.

- **Focus**
 - Ensure a workload performs its intended function correctly and consistently when it's expected to.
- **Key topics**
 - Designing distributed systems
 - Recovery planning
 - Handling change

Reliability design principles

Reliability pillar



Recover from
failure and
mitigate
disruption.

- Automatically recover from failure
- Test recovery procedures
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity
- Manage change in automation



Reliability questions

Foundations

- How do you manage service quotas and constraints?
- How do you plan your network topology?

Workload architecture

- How do you design your workload service architecture?
- How do you design interactions in a distributed system to prevent failure?
- How do you design interactions in a distributed system to mitigate or withstand failures?

Change management

- How do you monitor workload resources?
- How do you design your workload to adapt to changes in demand?
- How do you implement change?

Failure management

- How do you back up data?
- How do you use fault isolation to protect your workload?
- How do you design your workload to withstand component failures?
- How do you test reliability?
- How do you plan for disaster recovery?



Performance Efficiency pillar

Performance Efficiency pillar – use resources sparingly

Performance Efficiency pillar



Use
resources
sparingly.

- **Focus**
 - Use IT and computing resources efficiently to meet system requirements and to maintain that efficiency as demand changes and technologies evolve.
- **Key topics**
 - Selecting the right resource types and sizes based on workload requirements
 - Monitoring performance
 - Making informed decisions to maintain efficiency as business needs evolve

Performance efficiency design principles

Performance Efficiency pillar



Use
resources
sparingly.

- Democratize advanced technologies
- Go global in minutes
- Use serverless architectures
- Experiment more often
- Consider mechanical sympathy



Performance efficiency questions

Selection

- How do you select the best performing architecture?
- How do you select your compute solution?
- How do you select your storage solution?
- How do you select your database solution?
- How do you configure your networking solution?

Review

- How do you evolve your workload to take advantage of new releases?

Monitoring

- How do you monitor your resources to ensure they are performing?

Tradeoffs

- How do you use tradeoffs to improve performance?



Cost Optimization pillar

Cost Optimization pillar – eliminate unneeded expense

Cost Optimization pillar



Eliminate
unneeded
expense.

- **Focus**
 - Avoid unnecessary costs.
- **Key topics**
 - Understanding and controlling where money is being spent
 - Selecting the most appropriate and right number of resource types
 - Analyzing spend over time
 - Scaling to meeting business needs without overspending

Cost optimization design principles

Cost Optimization pillar



Eliminate
unnecessary
expense.

- Implement Cloud Financial Management
- Adopt a consumption model
- Measure overall efficiency
- Stop spending money on undifferentiated heavy lifting
- Analyze and attribute expenditure



Cost optimization questions

Practice cloud financial management

- How do you implement cloud financial management?

Expenditure and usage awareness

- How do you govern usage?
- How do you monitor usage and cost?
- How do you decommission resources?

Cost-effective resources

- How do you evaluate cost when you select services?
- How do you meet cost targets when you select resource type, size, and number?
- How do you use pricing models to reduce cost?
- How do you plan for data transfer changes?

Manage demand and supply resources

- How do you manage demand and supply resources?

Optimize over time

- How do you evaluate new services?



The AWS Well-Architected Tool

- Helps you review the state of your workloads and compares them to the latest AWS architectural best practices
- Gives you access to knowledge and best practices used by AWS architects, whenever you need it
- Delivers an action plan with step-by-step guidance on how to build better workloads for the cloud
- Provides a consistent process for you to review and measure your cloud architectures

Section 1 key takeaways



- The AWS Well-Architected Framework provides a **consistent approach** to evaluate cloud architectures **and guidance** to help implement designs.
- The AWS Well-Architected Framework documents a **set of design principles** and **best practices** that enable you to understand if a specific architecture aligns well with cloud best practices.
- The AWS Well-Architected Framework is organized into **six pillars**.
- Each pillar includes its own set of **design principles** and **best practices**.



Section 2: Reliability and availability

Module 9: Cloud Architecture

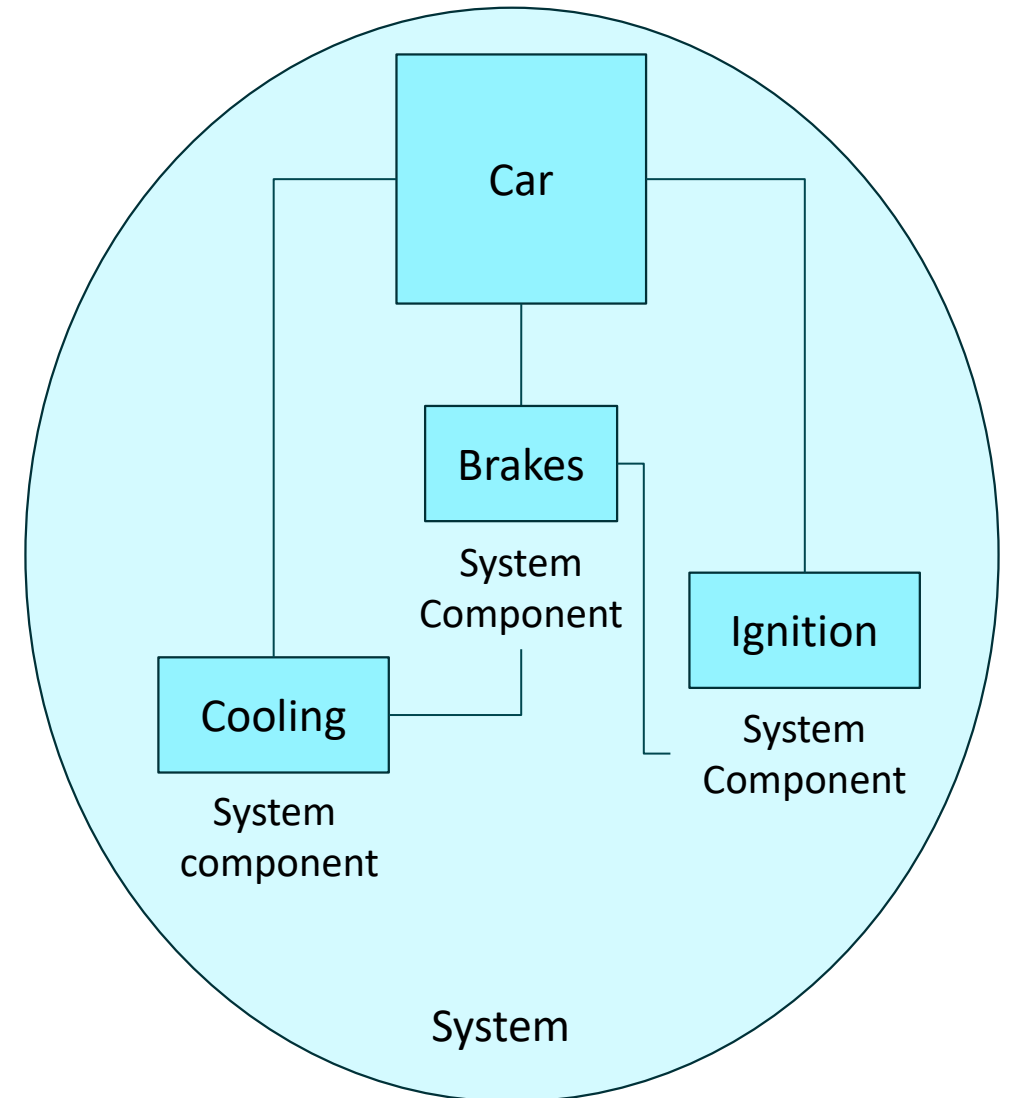


“Everything fails, all the time.”

Werner Vogels, CTO, Amazon.com

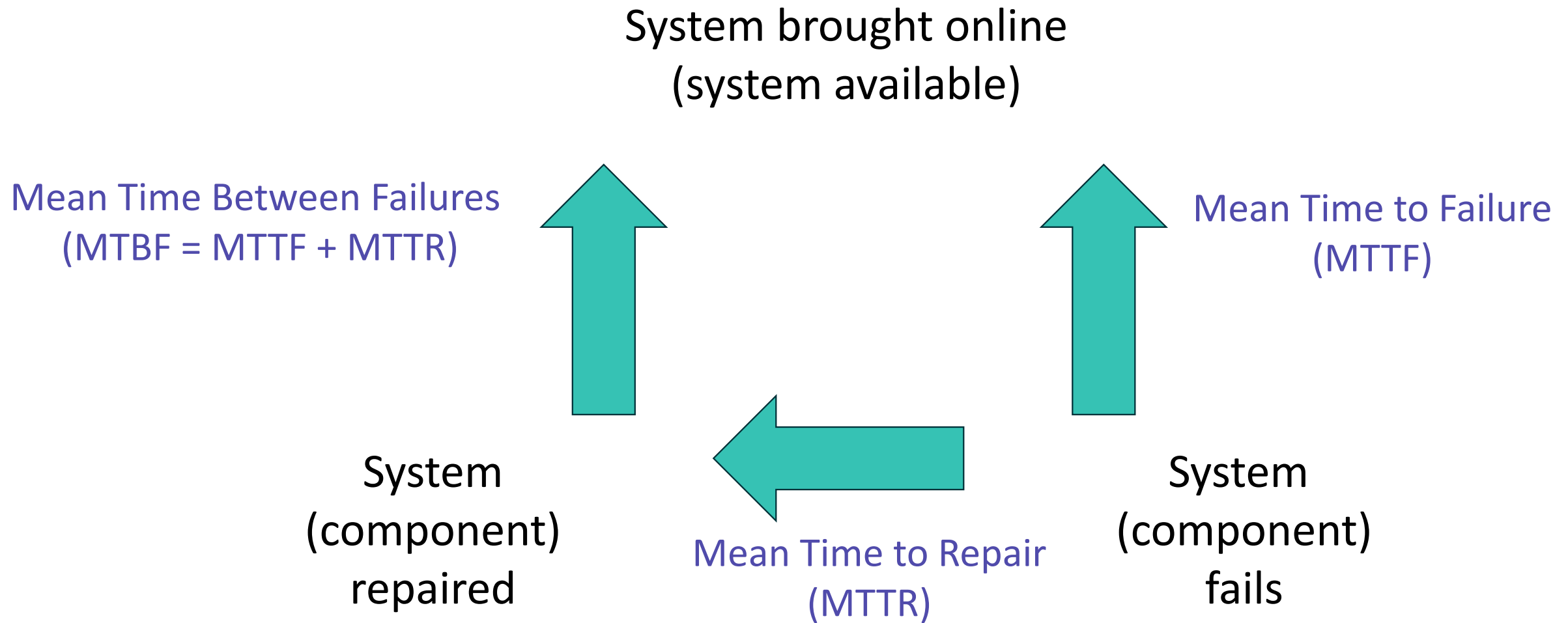
Reliability

- A measure of your system's **ability to provide functionality** when desired by the user.
- **System** includes all system components: hardware, firmware, and software.
- **Probability** that your entire system will function as intended for a specified period.
- **Mean time between failures (MTBF)** = total time in service/number of failures





Understanding reliability metrics





Availability

- Normal operation time / total time
- A percentage of uptime (for example, 99.9 percent) over time (for example, 1 year)
- Number of 9s – Five 9s means 99.999 percent availability



High availability

- System can withstand some measure of degradation while still remaining available.
- Downtime is minimized.
- Minimal human intervention is required.





Availability tiers

Availability	Max Disruption (per year)	Application Category
99%	3 days 15 hours	Batch processing, data extraction, transfer, and load jobs
99.9%	8 hours 45 minutes	Internal tools like knowledge management, project tracking
99.95%	4 hours 22 minutes	Online commerce, point of sale
99.99%	52 minutes	Video delivery, broadcast systems
99.999%	5 minutes	ATM transactions, telecommunications systems



Factors that influence availability

Fault tolerance

- The **built-in redundancy** of an application's components and its **ability to remain operational**.

Scalability

- The ability of an application to **accommodate increases in capacity needs** without changing design.

Recoverability

- The process, policies, and procedures that are related to **restoring service** after a catastrophic event.

Section 2 key takeaways



- **Reliability** is a measure of your system's ability to provide functionality when desired by the user, and it can be measured in terms of MTBF.
- **Availability** is the percentage of time that a system is operating normally or correctly performing the operations expected of it (or normal operation time over total time).
- Three factors that influence the availability of your applications are **fault tolerance**, **scalability**, and **recoverability**.
- You can design your workloads and applications to be **highly available**, but there is a cost tradeoff to consider.



Section 3: AWS Trusted Advisor

Module 9: Cloud Architecture

AWS Trusted Advisor



AWS Trusted Advisor

- Online tool that provides real-time guidance to help you provision your resources following AWS best practices.
- Looks at your **entire AWS environment** and gives you real-time recommendations in five categories.

Cost Optimization



0 ✓ 9 ⚠ 0 !

\$7,516.85

Potential monthly savings

Performance



3 ✓ 7 ⚠ 0 !

Security



2 ✓ 4 ⚠ 11 !

Fault Tolerance



0 ✓ 15 ⚠ 5 !

Service Limits



37 ✓ 0 ⚠ 1 !

Section 3 key takeaways



- [AWS Trusted Advisor](#) is an online tool that provides real-time guidance to help you provision your resources by following AWS best practices.
- AWS Trusted Advisor looks at your [entire AWS environment](#) and gives you real-time recommendations in five categories.
- You can use AWS Trusted Advisor to help you optimize your AWS environment as soon as you start implementing your architecture designs.



Module wrap-up

Module 9: Cloud Architecture



Module summary

In summary, in this module you learned how to:

- Describe the AWS Well-Architected Framework, including the six pillars
- Identify the design principles of the AWS Well-Architected Framework
- Explain the importance of reliability and high availability
- Identify how AWS Trusted Advisor helps customers
- Interpret AWS Trusted Advisor recommendations

Complete the knowledge check





Sample exam question

A SysOps engineer working at a company wants to protect their data in transit and at rest. What services could they use to protect their data?

Choice	Response
A	Elastic Load Balancing
B	Amazon Elastic Block Storage (Amazon EBS)
C	Amazon Simple Storage Service (Amazon S3)
D	All of the above



Additional resources

- AWS Well-Architected website: <https://aws.amazon.com/architecture/well-architected/?wa-lens-whitepapers.sort-by=item.additionalFields.sortDate&wa-lens-whitepapers.sort-order=desc>
- AWS Well-Architected Labs: <https://wellarchitectedlabs.com/>
- AWS Trusted Advisor Best Practice Checks: <https://docs.aws.amazon.com/awssupport/latest/user/trusted-advisor-check-reference.html>



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